Edward Ajayi

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Education

Carnegie Mellon University
Master of Science in Engineering Artificial Intelligence — *Aug. 2024 – Present*
- **Relevant Courses:** Natural Language Processing, Deep Learning, Question Answering with LLMs
- **Specializations:** Data Science, Machine Learning, NLP
- **GPA:** 3.77/4.0

Ekiti State University
Bachelor of Science in Computer Science — *Oct. 2014 – Nov. 2019*
- **Specializations:** Computer Science
- **GPA:** 4.17/5.0

Research Interests

Language Modelling, Representation Learning, Socially Aware AI, Explainable AI, Low-Resource NLP

Selected Publications

1. **E. Ajayi**, M. Kachweka, M. Deku, and E. Aiken. *A Machine Learning Approach for Detection of Mental Health Conditions and Cyberbullying from Social Media. Proceedings of Machine Learning Research (PMLR), AAAI-26 AI in Medicine and Healthcare Bridge*. **Best Paper Award**. 2026. [PMLR]
2. **E. Ajayi** and P. Mitra. *HumorRank: An Automated Leaderboard for Evaluating Humor Generation in Large Language Models.* Manuscript in preparation for *COLM*. 2026. [Manuscript]
3. **E. Ajayi** and P. Mitra. *HumorGen: Cognitive Synergy for LLM Humor Generation via Persona-Based Distillation.* Submitted to *EMNLP*. 2026. [Manuscript]
4. **E. Ajayi** and P. Mitra. *Automatic Humor Detection: From Theoretical Foundations to Large Language Models.* Submitted to *ACM Computing Surveys*. 2025. [Manuscript]
5. **E. Ajayi**, B. Tadele, E. Umwari, M. Deku, P. Singadi, C. Edeh, and J. Udahemuka. *Language Diversity: Evaluating AI Performance on African Languages.* arXiv Preprint. 2025. [arXiv Preprint]
6. **E. Ajayi**. *AfriEconQA: A Benchmark Dataset for African Economic Analysis Based on World Bank Reports.* Under Review at ACM SIGIR. 2026. [Manuscript]

Experience

Predoctoral Research Fellow
Human-Algorithm Interaction Lab, Oxford Saïd Business School, University of Oxford, UK — *Feb. 2026 – Present*
- Conduct research on Human-AI collaboration, decision-making, and AI adoption in service systems at the Human-Algorithm Interaction Lab.

Research Assistant
Spatial and Language Technologies Lab, CMU — *Aug. 2025 – Present*
- Conducting research on NLP for automatic humour detection and generation, dataset preparation, model development, and evaluation.
- Contributing to projects on low-resource language technologies and advancing cultural understanding in large language models.

Graduate Researcher
Carnegie Mellon University — *Jan. 2025 – Present*
- Researched machine learning methods for detecting mental health conditions and cyberbullying from social media, resulting in a paper accepted at AAAI-26.
- Conducting two ongoing research projects in Question Answering with Large Language Models, focusing on reasoning, evaluation, and robustness.

Research Intern - Business Automation
Carnegie Mellon University — *Jun. 2025 – Aug. 2025*
- Developing and prototyping an agentic AI framework to automate complex business workflows.
- Researching and implementing AI safety protocols and robust LLM reasoning for ethical alignment and deployment.
- Analyzing agent performance to establish benchmarks for human-in-the-loop oversight and develop explainability metrics to support transparent corporate accountability.

Client Service Officer
Stanbic IBTC Bank — *Dec. 2021 – Jul. 2024*
- Leveraged data-driven insights to cross-sell over N400M in assets, enhancing client trust and digital adoption by 25%.
- Managed client relationships, identifying service gaps to improve operational efficiency.

Technical Skills

Languages: Python, SQL, Bash
Generative AI & LLMs: LangGraph, OpenAI Agents SDK, RAG, LLM Fine-Tuning, Prompt Engineering, DPO
Deep Learning & ML: PyTorch, Hugging Face Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, Seaborn
Natural Language Processing: spaCy, NLTK, Sentiment Analysis, Text Processing
Data Engineering & Ops: Docker, AWS, FastAPI, Neo4j, BeautifulSoup, Jupyter

Projects

HumorGen: Cognitive Synergy for Constrained Humor Generation | *Qwen-7B, DPO, Distillation*
- Developed the Cognitive Synergy Framework (CSF), a theory-driven approach using cognitive personas to generate diverse humor reasoning patterns.
- Developed **HumorGen**, a 7B model achieving performance competitive with state-of-the-art models on humor generation tasks. HuggingFace

Multilabel Machine Learning System for Mental Health and Cyberbullying Detection
- Built a full ML pipeline for multilabel detection of mental health conditions and cyberbullying using classical models and transformer-based embeddings.
- Produced results leading to an *Oral* paper acceptance at AAAI-26 AIMedHealth. [arXiv] GitHub

AfriEcon-QA: Hybrid RAG Benchmark for Economic QA | *Python, LangChain, RRF*
- Developed **AfriEcon-QA**, a closed-domain QA benchmark for economic reports, validating the necessity of a **Hybrid Reciprocal Rank Fusion (RRF)** strategy for precision.
- Analyzed LLM scaling limits, identifying context instability in open-weight models (Qwen-2.5) under deep retrieval, informing robust RAG deployment.

MyTorch – A Minimal Deep Learning Framework Built from Scratch | *Python*
- Implemented core tensor ops, autograd, activation layers, and SGD without deep learning libraries.
- Designed a PyTorch-style API enabling end-to-end training and illustrating neural network internals.

AI Workflow Automation Agent | *Python, LangChain, FastAPI, Google Sheets API, SMTP*
- Built an autonomous agent that processes Google Form submissions, categorizes requests, and triggers approval workflows.
- Automated notifications and integrated a QA agent that validates updates and writes results in real time.

Training & Conferences

Oral & Poster Presentation, AAAI Conference on Artificial Intelligence — Singapore
AAAI — *Jan. 2026*
- Presented my paper on ML for mental health and cyberbullying detection from social media, and engaged with academics and researchers at the conference.

Participant, Deep Learning Indaba 2025 — Rwanda
Deep Learning Indaba — *Aug. 2025*
- Attended workshops on deep learning, AI for social good, and research collaboration across African universities.

Participant, Digital Humanities Summer School — Rwanda
DIGIHUM/CMU-Africa — *Jul. 2025*
- Engaged in computational methods for humanities research, including NLP applications.

Volunteering/Leadership

President, CMU-Africa Research Club
Carnegie Mellon University — *Jan. 2025 – Dec. 2025*
- Led 12 major events in 2025 supporting student research.
- Launched website with research guides, conference updates, and collaboration hub.
- Organized first Student Research Showcase with 20+ posters and workshops.
- Hosted three-part PhD Application Series, faculty sessions, and alumni panels supporting 40+ students.
- Collaborated with ACM SIGCOMM and Upanzi Network for remote conference participation.
- Built an active research ecosystem that provided critical peer support, contributing to over 20 student acceptances to top-tier venues including NeurIPS and AAAI in 2025.

Reviewer, AAAI AIMedHealth Bridge 2026
AI for Medicine and Healthcare — *Nov. 2025 – Nov 2025*
- Reviewed submissions in applied machine learning and AI for healthcare under the AAAI AIMedHealth Bridge 2026 program.

IEEE Student Member, CMU-Africa
Institute of Electrical and Electronics Engineers — *Feb. 2025 – Present*
- Collaborating with IEEE members to advance research initiatives among club members.

Certifications

- Microsoft Certified Azure Data Scientist Associate - View
- DataCamp Python Data Associate - View

Honors and Awards

- Mastercard Foundation Scholarship, Carnegie Mellon University Africa
- Federal Scholarship Board Award for Undergraduates, Ekiti State University